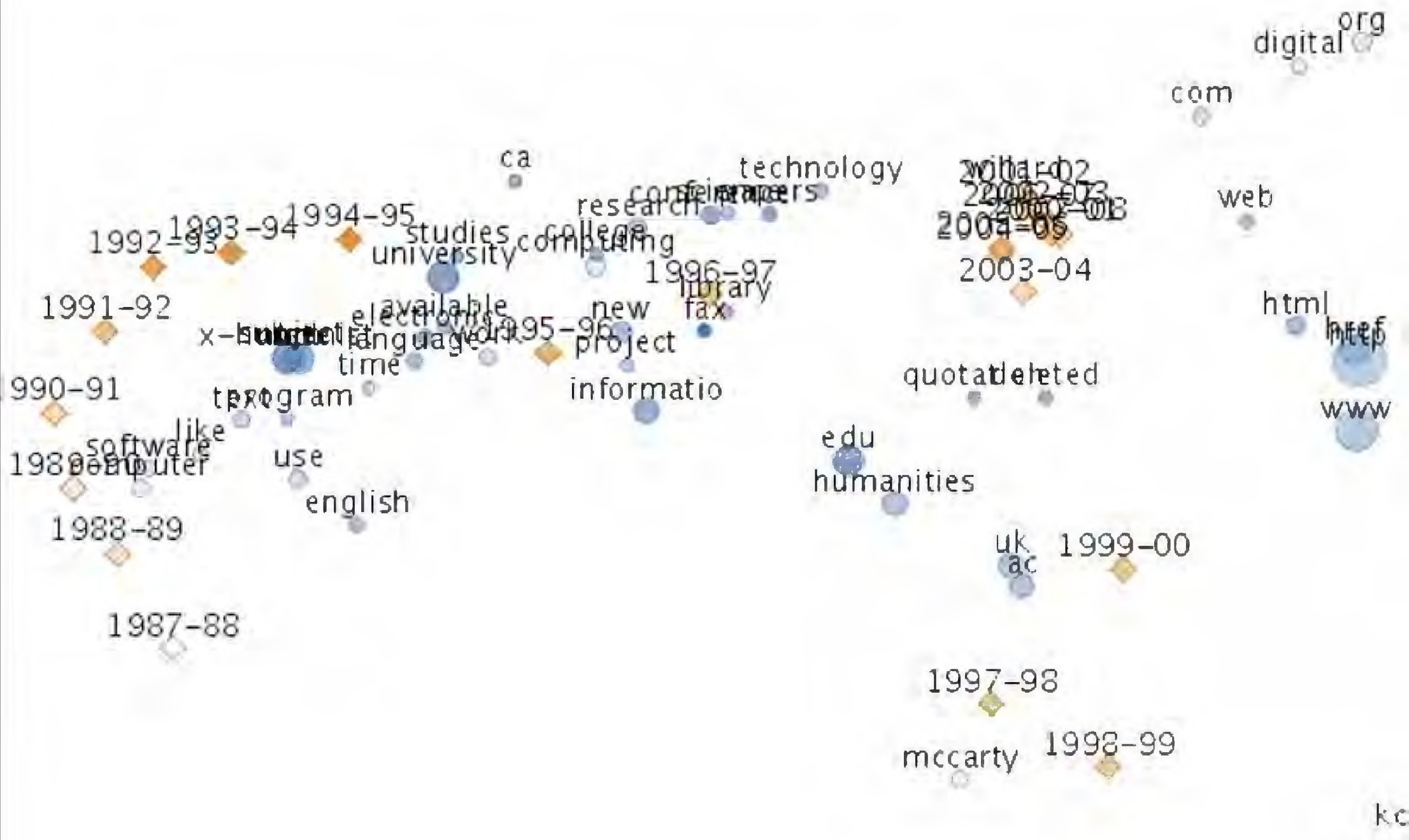



# HERMENEUTICA

## Computer-Assisted Interpretation in the Humanities

Geoffrey Rockwell and Stéfan Sinclair

# Hermeneutica

# Hermeneutica

## Computer-Assisted Interpretation in the Humanities

Geoffrey Rockwell and Stéfan Sinclair

The MIT Press
Cambridge, Massachusetts
London, England

© 2016 Massachusetts Institute of Technology

All rights reserved. No part of this book may be reproduced in any form by any electronic or mechanical means (including photocopying, recording, or information storage and retrieval) without permission in writing from the publisher.

Set in Stone Serif Std by Toppan Best-set Premedia Limited. Printed and bound in the United States of America.

Cataloging-in-Publication information is available from the Library of Congress.

ISBN: 978-0-262-03435-7

10 9 8 7 6 5 4 3 2 1

# Contents

Acknowledgments vii

1 Introduction: Correcting Method 1

2 The Measured Words: How Computers Analyze Texts 25

3 From the Concordance to Ubiquitous Analytics 45

4 The Swallow Flies Swiftly Through: An Analysis of Humanist (First Interlude) 69

5 There's a Toy in My Essay: Problems with the Rhetoric of Text Analysis 83

6 Now Analyze That! Comparing Two Speeches on Race (Second Interlude) 105

7 False Positives: Opportunities and Dangers in Big-Data Text Analysis 113

8 Name Games: Analyzing *Game Studies* (Third Interlude) 137

9 A Model Theory: Thinking Through Hermeneutical Things 149

10 The Artifice of Dialogue: Thinking Through Scepticism in Hume's *Dialogues* (Fourth Interlude) 167

11 Agile Hermeneutics and the Conversation of the Humanities 191

Notes 205

Bibliography 227

Index 241

# 7 False Positives: Opportunities and Dangers in Big-Data Text Analysis[1]

> The NSA has built an infrastructure that allows it to intercept almost everything. With this capability, the vast majority of human communications are automatically ingested without targeting. If I wanted to see your emails or your wife's phone, all I have to do is use intercepts. I can get your emails, passwords, phone records, credit cards. I don't want to live in a society that does these sort of things. … I do not want to live in a world where everything I do and say is recorded. That is not something I am willing to support or live under.
>
> Edward Snowden, June 10, 2013 (quoted in MacAskill 2013)

In May of 2013, Edward Snowden, a former security contractor for the US government, flew to Hong Kong with a large cache of documents about the National Security Agency. He shared these materials with news organizations. Some of those organizations, including the *Guardian*, have since been carefully preparing stories based on the documents, stories that reveal something of the extent and sophistication of the surveillance conducted by the NSA and other intelligence outfits.[2] One set of documents revealed the existence of PRISM and Boundless Informant, software that let analysts search the metadata and even the full texts of messages to and from US citizens. Text analysis on a very large scale is clearly happening, and analytical tools are being developed by governments and corporations; they are spending millions if not billions of dollars on developing infrastructure and tools for analyzing and storing metadata and text.[3] The scale of the infrastructure and how it is being used raise challenging social and ethical issues that force us all, as citizens in a digital society, to confront how we balance priorities. As Jeffrey Rosen observes, the debate is, in part, about the balance between privacy and security:

> We run the risk, therefore, of constructing vast but ineffective architectures of surveillance and identification that threaten the liberty and privacy of innocent citizens without protecting us from terrorism.[4]

Does all this investment in surveillance and data tracking work? One of Liberace's sayings was "Too much of a good thing is wonderful." That, alas, may not be true of signals intelligence, which is why in this chapter we look at the phenomenon called "big data" and at what it means to practices of interpretation in the humanities, a domain in which we have traditionally assumed that more information is better.

Why does big data matter to the humanities? It matters because we—as a society and as academics—have the fever for ever larger amounts of information. It matters because there is much at stake, from political and economic realities to a reconfiguration of how we know and study human history and culture (and ourselves) on a large scale. With the shift from the interpretation of carefully chosen examples (of literary texts, historical documents, philosophical dialogues, and so on) to the analysis of datasets that are beyond our capacity to read individually, there are potentially significant transformations in what we study and how we study. There are also new ethical dilemmas having to do with privacy. Individuals can be identified in big data in a way that they can't as easily be identified in carefully sampled and curated data. It has become hard to anonymize data when there are so many sources of data which, when cross-linked, can reveal identity.[5] Further, there are issues around the persistence of data (such as questionable pictures posted to a social network or stored in the cloud) that may be impossible to erase from view.[6] Beyond issues of privacy, interpretation changes when there is a shift from studying unique works of art through practices that privilege close reading to the exploration practices of large-scale distant reading.

This chapter is not intended to explain the technical aspects of big data or the opportunities it presents. (Plenty of books, articles, websites, and videos are available for those purposes.) Here we will limit ourselves to a brief definition of big data and will focus on how understanding big data is important to understanding the opportunities for textual interpretation in the humanities. We will survey the phenomenon of big data in order to understand any implications for the future of computer-assisted text analysis and hermeneutica in the humanities.

In particular, we will use examples related to national security organizations such as the NSA and the CSEC (Communications Security Establishment Canada) to examine big data. We will not discuss exactly what our governments are gathering or what they are doing with what they gather. Thankfully, the *Guardian*, the *Washington Post*, the *Intercept*, the *New York Times*, the Electronic Frontier Foundation, and the Electronic Privacy Information Center are investigating their activities, and by the time you read

this there will have been more revelations from Snowden's hoard.[7] Instead we will use what has been discovered about government projects, both older and recent, to illustrate what *can* be done with big-data text analysis. By looking away from literary text analysis to what agencies with lots of resources can do with analytics, we can better see the dangers and opportunities. Moreover, there are striking similarities in the search for unusual phenomena within a large collection of literary texts that merit further interpretation and the search for suspicious communication within a global network that warrants closer attention. Both activities are essentially about calibrating signal-to-noise ratios where defining what is a signal for computational processing is a deceptively difficult operation.

## What Is Big Data?

With their characteristic ability to provide memorable formulations, the business and IT sectors have latched on to a set of words beginning with the letter V to describe big data. The original formulation may have only contained Volume, Velocity, and Variety, but several variants add Veracity and Value.[8] The central idea is that, owing to changes in how information is created, stored, and transmitted, there is now a lot (Volume) of heterogeneous data (Variety), sometimes coming in very quickly (Velocity), from which businesses or governments can extract new truths (Veracity) and make money (Value).

As the "big" in "big data" suggests, the volume of data is the novel factor, but that doesn't mean that there is a precise threshold of size. It is more a matter of the size provoking a change in how you handle and interpret the data. Once the amount of data gets beyond the point where traditional human forms of analysis work or traditional technologies work, you have big data. Google had to develop new techniques to handle the volume of data gathered and indexed from the World Wide Web so that they could let millions of people search trillions of pages. Going further than just handling lots of data, Google developed the PageRank algorithm for showing the pages most linked to among all the results. There is no point to providing users with a list of millions of pages that match the search criteria that they then have to page through; it is more useful to rank the pages so that the most relevant results appear on the first screens of the list. The simple concordance as a way of organizing results had to be updated when the quantity of data became so great that even the results of a search were too big to handle.

Likewise, an excess of textual data means that scholars can't read it all and have to switch to different methods in order to be able to synthesize insights from the whole. That is what Brunet was writing about in the 1980s when he was analyzing the *grand corpus* of FRANTEXT and what Franco Moretti discussed more recently in *Graphs, Maps, Trees*. Moretti coined the term "distant reading" to distinguish what he was doing from "close reading," but also to suggest metaphorically that with large quantities of data we have to step back to get a view of the whole. When the quantity of data is too big for humans to read, different practices are needed to understand the whole. That change in practices also changes the *what* and the *how* you interpret. Insofar as a theory is a view of the whole, with big data one needs a different kind of theoretical gaze, not merely a scaled-up one. Some would suggest that it isn't a theory at all when you let the data drive science, and when huge amounts of data make it practicable to compute correlations without prior hypotheses or theoretical models (Anderson 2008).

Another way to put this is that interpretation undergoes a qualitative change when the amount of information gets so large that no one can read it all. When that happens, you switch to sampling the phenomenon (as is done in the social sciences and polling), or you closely read selections as representatives of the whole (as is done in the humanities), or you have to find ways to handle all the information—and that involves some form of automatic data processing.[9] As Moretti points out in a collection of essays titled *Distant Reading*, when you stand back and use a computer to read data you are no longer doing interpretation as interpretation is traditionally understood; you are explaining. Interpretation engages the meaning of the text. It brings forth the idiosyncratic uniqueness. Explaining, on the other hand, shows the relationships between documents and can be used to trace a history of relationships. Explaining is not about re-evoking the meaning of a literary text, but about discovering the large-scale patterns of change in texts. It is epidemiology; it tracks symptoms, not causes.

Moretti goes on to assert that we don't need any more literary interpretation, as we have more than enough of it.[10] Here we side with Liberace; while there may be too much interpretation, it's a good and human thing, and that's the wonder. Like breathing, interpreting is such a fundamentally human activity that we cannot stop no matter how much of it there is. Of course Moretti isn't talking about the everyday interpreting we all do, but about the institutional investment in academic interpretation.

Big data is more than just about volume, but also a change in technique brought on and made possible by scale. When you have all or much of the available data rather than only samples, you can ask different types of

questions. As we pointed out about hermeneutica in chapter 6, you can even move from distant reading to interpretive detail and then back to distant reading. That is what the NSA wants to do, though for surveillance and predictive purposes. They don't just want samples in order to understand terrorism, and they don't just want to be able to track any identified target's communications. They are preserving data so they will be able to backtrack. They are also experimenting with whether they can identify new suspects through sophisticated profiling, which slides into trying to predict who might "radicalize." For these new approaches, the NSA needs all the data it can get—more than just the names of identified suspects. It wants to work at multiple levels. Being limited to sampling would constrain the NSA to levels it had received clearance for and would provide only a partial view based on the samples collected.

The value of big data lies partly in the secondary and unanticipated uses. Humanists are familiar with the desire to have everything at hand so as to be able to explore new questions later (without needing to measure and collect new data) or to serendipitously follow any interpretive path, even one not anticipated by the author of a literary text. The bigger the library, the more you have ready to hand as fodder for interpretation.

## Metadata and Content

Another way to understand big data is to understand some of the types of data that can be collected and analyzed. One type, made famous by the first of the Snowden leaks, is metadata. In defending NSA practices, Senator Dianne Feinstein describes telephone metadata as follows:

> The call-records program is not surveillance. It does not collect the content of any communication, nor do the records include names or locations. The NSA only collects the type of information found on a telephone bill: phone numbers of calls placed and received, the time of the calls and duration. The Supreme Court has held this "metadata" is not protected under the Fourth Amendment.[11]

On June 6, 2013, the *Guardian* published an article on the first document leaked by Snowden: a court order that compelled the telephone-service provider Verizon to hand over to the NSA "all call detail records" about customers' calls.[12] This leak and the various responses, including Senator Feinstein's response quoted above, introduced into common parlance the term "metadata," meaning "data about data." The metadata about a phone call includes the date of the call, the duration, the number called, and the location of the caller. Metadata doesn't include the substance of the

call, but with enough metadata one can learn a lot about someone. With enough metadata, one can chain information to plot the network of connections between people, places, and organizations.

Metadata has the added virtue that it is more efficient than the full contents of calls. The problem with big data is that there is too much of it, and that is even more so for images, audio, video, and other storage-intensive multimedia formats. In many cases we have convenient metadata that provides a surrogate for what is most important about a phenomenon. The phone companies, for example, keep metadata about calls that can be analyzed without trying to capture and analyze the full audio stream of billions of calls. Likewise, in the humanities we have bibliographic databases, such as the MLA International Bibliography, that contain carefully curated metadata about publications that can now be exploited as big data. As Matthew Jockers puts it in his 2013 book *Macroanalysis*, this "type of catalog metadata has been largely untapped as a means of exploring literary history,"[13] though the notion of studying literature through card catalogs may seem blasphemous to some.

It isn't always easy to tell the difference between metadata and content, especially as our analytical tools can treat metadata as content or use full-text to generate metadata. Is the title of a work part of the content, or is it metadata? Franco Moretti (2013b) talks about networks of characters in Shakespeare plays. For his visualizations he links characters when they talk to one another. The character/link data he draws from the text is metadata drawn from text data and then visualized. This sounds a lot like the building up of layers of structure that John B. Smith discussed in "Computer Criticism" (1978). Moretti makes a further interesting point about working with such metadata: "once you make a network of a play, you stop working on the play proper, and work on a *model* instead. You reduce the text to characters and interactions, abstract them from everything else."[14] It is surrogates all the way down, but some surrogates are easier to handle. Metadata is often more efficient, and its analysis can show you things that you couldn't see easily in the full haystack, but it involves an abstraction that can distorts as it hides and reveals.

What shouldn't surprise us is that one of the main forms of metadata we like to extract is metadata about people and how they are connected—the proliferation of tools to explore networks of friends on Facebook is a manifestation of this in popular culture. A more covert example is the NSA's program NYMROD, which sifts through massive databases to collate reports on people the NSA wants to track, such as Angela Merkel (Poitras et al. 2014). Others have imagined similar surveillance tools for

the humanities. In "What Do You Do with a Million Books?" Greg Crane (2006) calls for "the ability to extract from the stored record of humanity useful information in an actionable format for any given human being of any culture at any time." Crane and others are building the technological means for extracting large numbers of structured facts (linked data) about people, human history, and culture by finding patterns, such as [person] was born at [place]. IBM's Watson takes this further, extracting data so as to perform deep analysis (and association—or networking—of people, places, locations, events, and other data) in order to enable open-ended question answering using natural language, as was demonstrated by a mock Jeopardy game (Ferruci et al. 2011).

## Information in Motion and Information at Rest

Another distinction that we can see in the Snowden leaks is between the fairly stable information in databases that gets called "information at rest" and the data flowing through various communication channels, which gets called "information in motion."[15]

Information at rest is what we usually think of as big data and is more commonly used for data mining than information in motion is. It consists of all the large databases of information about people and things. It is not a flow of data, but the structured and curated data gathered over the years that can be mined for insight. At any large university there are a number of such large databases, including personnel and student databases and library circulation databases. Businesses and government likewise maintain large databases, which if linked and mined can generate new insights.

From what we can tell about the NSA's PRISM project from the slides leaked, PRISM is an Information at Rest system that can search large corporate databases. The *Washington Post* has described PRISM as a tool that lets users initiate search requests, or "tasks," that go to corporate databases. According to the *Post*, "depending on the company, a tasking may return e-mails, attachments, address books, calendars, files stored in the cloud, text or audio or video chats and 'metadata' that identify the locations, devices used and other information about a target."[16]

We are all collaborators in many of these really large databases. If you use one of the U.S. service providers to which PRISM has access, your data could be "tasked." If you use a loyalty card that gets you discounts at your local supermarket, you are contributing to a commercial database that gathers information about purchasing habits which can be correlated with the personal information you provided to the supermarket and with your credit

card information. If you upload a video to YouTube, you are contributing to a massive multimedia database. If you have a Facebook account, you are contributing to a huge "social graph" of who is "friends" with whom, what they say they like, what they say they are doing, and so on. It is estimated that in 2012 Facebook had a billion users—more than 10 percent of the world's population (Mayer-Schönberger and Cukier 2013, p. 92). That's a lot of information about a lot of us held by one corporation, and susceptible to access by a government.

What can be done with such data? A number of the uses are fairly benign. Modern consumers probably already have a sense of how stores are using data to put impulse-bought products near checkout lines. Another use of such data is "market basket" analysis: the study of what products are in the same "basket" (or shopping cart) of buyers at a checkout. In the well-known "beer and diapers" story, data mining is supposed to have shown that on Friday and Saturday evenings many shopping carts in certain shopping centers contained beer and diapers, a combination that otherwise seems strange (Power 2002). The inference was that husbands were being sent to pick up diapers and were picking up some beer on impulse. The commercial advantage of such data mining is that a market can then locate beer near diapers, or between the diaper section and the checkout, in order to increase sales. A 2012 article in the *New York Times* describes similar data-mining efforts that the store chain Target used successfully to identify pregnant customers from their buying patterns so as to provide customized advertising[17]:

> As [the statistician Andrew] Pole's computers crawled through the data, he was able to identify about 25 products that, when analyzed together, allowed him to assign each shopper a "pregnancy prediction" score. More important, he could also estimate her due date to within a small window, so Target could send coupons timed to very specific stages of her pregnancy.[18]

With big data, organizations can make novel types of inferences and even predict things about their customers. However, it turns out, customers don't appreciate the attention. Women are understandably spooked when companies infer that they are pregnant. Big data is powerful in some ways but blunt in others, as evidenced by a woman who kept seeing advertising evolve throughout her presumed pregnancy and birth even though she had miscarried.[19]

Networked databases can be used to create composite pictures of individuals. Any one database may only contain a limited amount of data, but multiple databases can be mined to aggregate data (with all the attendant

perils of combining partial data sources). Software can connect a shopper's address, phone number, buying habits, credit information, and credit-card buying habits.[20]

Information in motion, by contrast, is the massive flow of phone conversations, e-mails, Web-page requests, and Internet transactions over fiber-optic cables, satellites, or local Wi-Fi. The "Upstream" system shown in the diagram reproduced here as figure 7.1 seems to be a system capable of tapping into and buffering the huge international flow of data. Upstream takes advantage of the fact that "much of the world's communications flow through the U.S." (as is explicitly indicated in the diagram reproduced here as figure 7.2).

How is the volume of information in transit studied? The *Independent Technical Review of the Carnivore System* Final Report of 2000, prepared by the IIT Research Institute, discusses how this was done by an earlier surveillance system called Carnivore. Essentially, Carnivore tapped into a network so as to be able, in theory, to listen in on all the traffic. It then filtered the flow

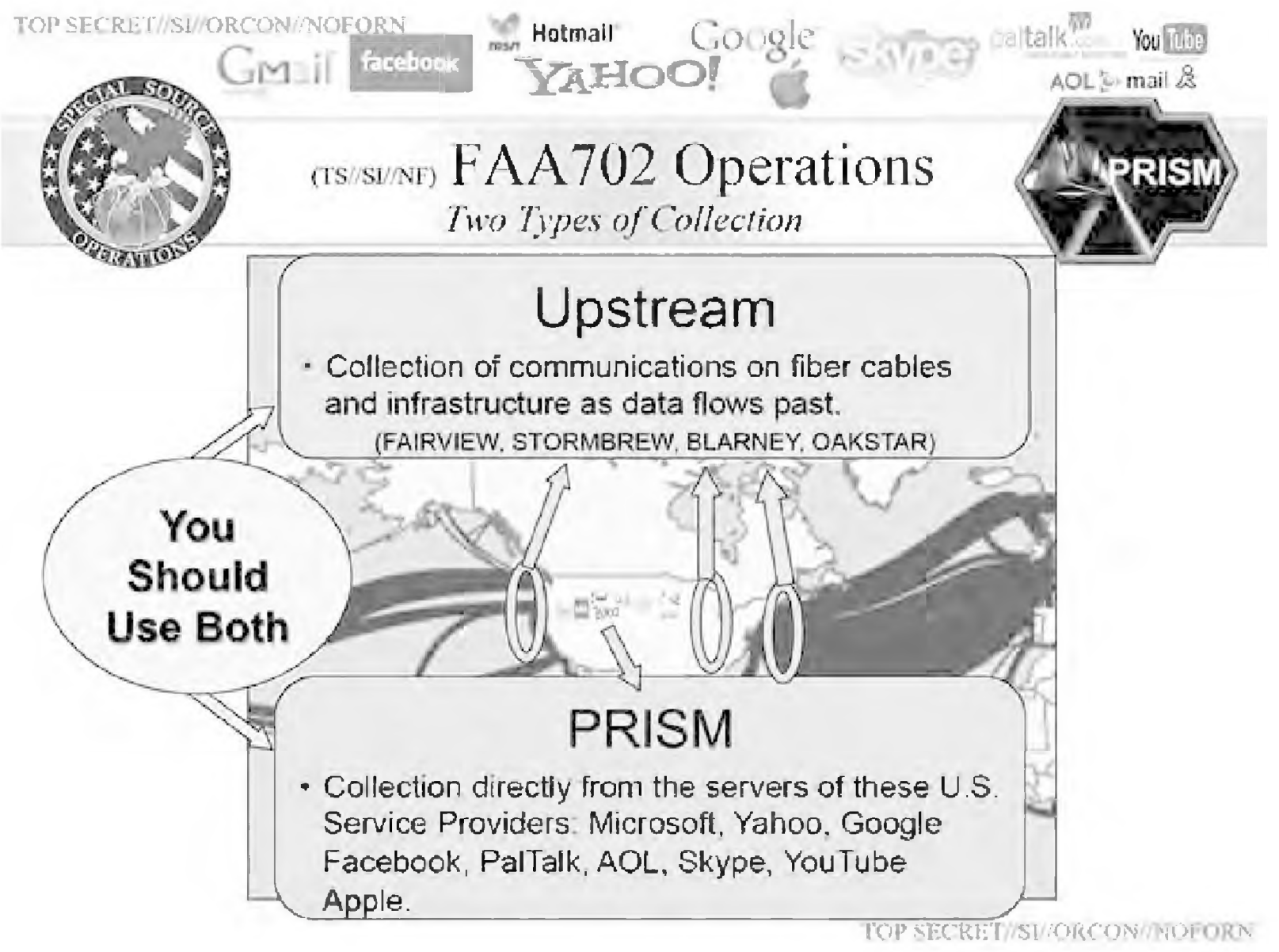



**Figure 7.1**

"Two Types of Collection" (a PRISM slide). Source: supplement to Gellman and Poitras 2013 titled "NSA slides explain the PRISM data-collection program" at http://www.washingtonpost.com/wp-srv/special/politics/prism-collection-documents.

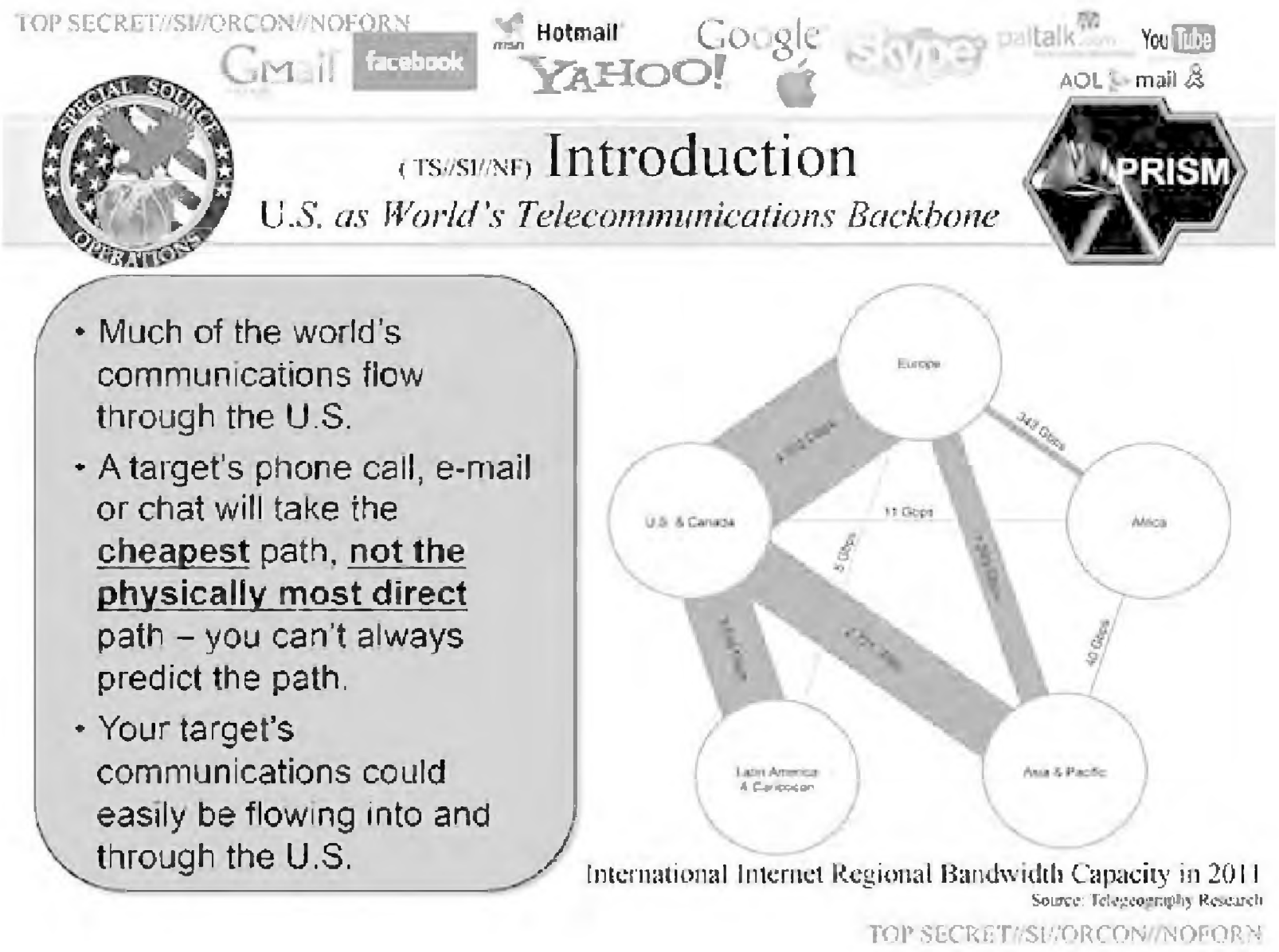



**Figure 7.2**
"U.S. as World's Telecommunications Backbone" (a PRISM slide). Source: "NSA slides explain the PRISM data-collection program" (http://www.washingtonpost.com/wp-srv/special/politics/prism-collection-documents/).

for information meeting various criteria. Because of the huge amount of information traveling on major trunk lines, specialized systems are needed to filter Information in Motion in real time (and it is worth reiterating that the rate at which we are producing more information is always increasing, roughly following Moore's Law by doubling every 18–24 months).[21] Systems such as Carnivore combine specialized signal-processing hardware that can handle huge flows of information with software that enables users to set up and manage the filters. Users can filter for lists of phone numbers, names, words, phrases, and addresses. These systems don't gather everything that is going by. The filters capture only the messages that match something analysts are interested in, such as the communications of particular people or communications about certain subjects. These messages or activities are then passed on to human analysts or stored so that they can be searched later. In combination with personal information gathered from databases, the communications information can be used to build up a "social graph" from all the metadata that is comparable to what Facebook

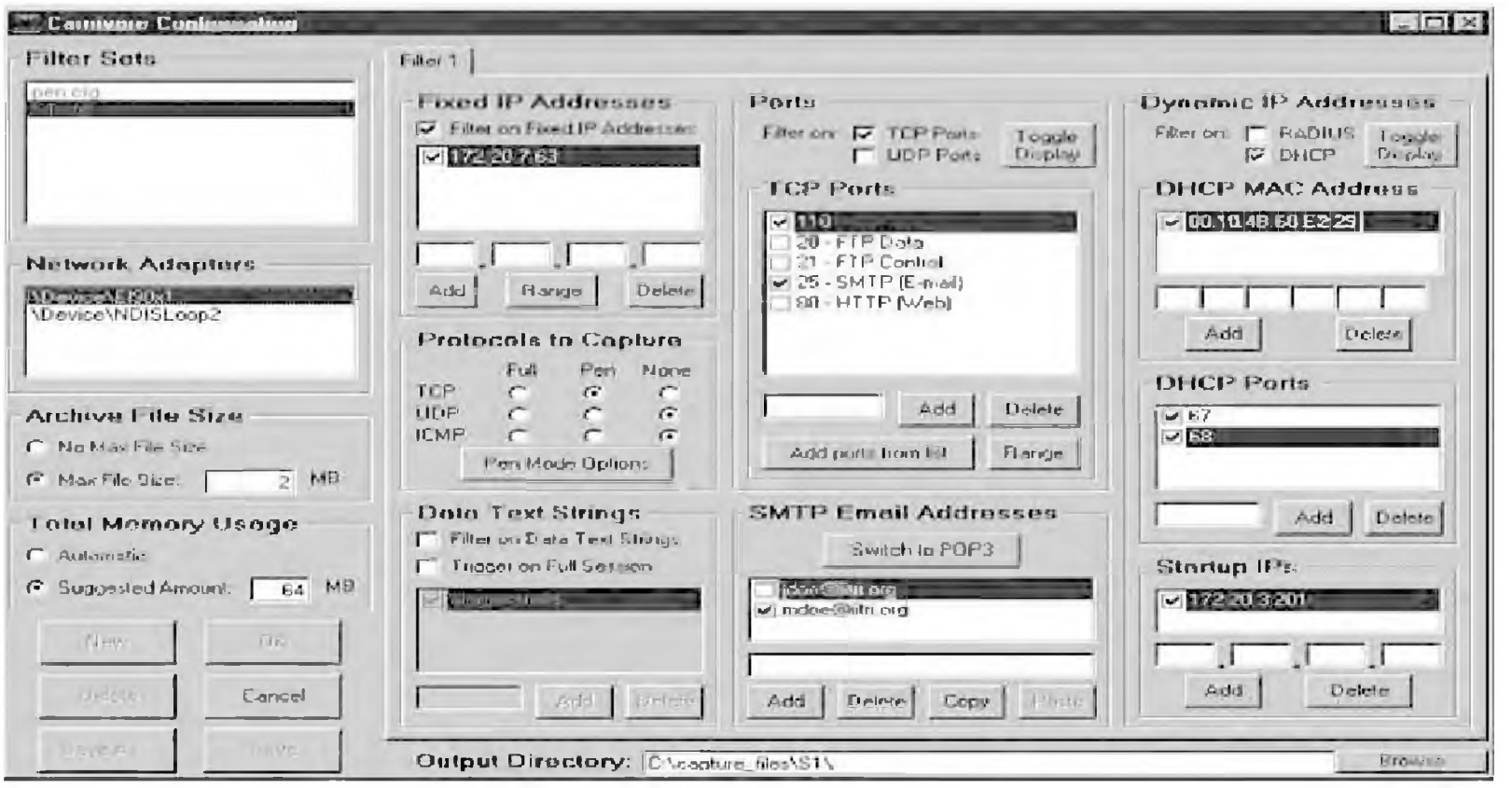



**Figure 7.3**
The interface used to set up a filter for Carnivore. (Carnivore is essentially an outdated system, but the panel conveys a sense of what "professional" Information in Motion gathering system can do.)

has, but with much more detail about who is connected with whom and what they are doing in real time. This can then be mined by means of social network analysis to follow networks of people, such as known terrorists or people communicating from suspect locations.

If you think these surveillance systems for information in motion are just for governments, think again. Narus sells a product, called Narus-Insight, that may be similar to what governments use to perform real-time semantic analysis. According to Narus' Product webpage,

> NarusInsight is Narus' flagship product and is the most scalable, real-time traffic intelligence system for the protection and management of large IP networks. It captures, analyzes and correlates IP traffic in real time and offers wide visibility across heterogeneous networks and deep insight into multiple layers of network traffic in real time. Through its patented analytics, NarusInsight's carrier-class software detects patterns and anomalies that can predict and identify security issues, misuse of network resources, suspicious or criminal activity, and other events that can compromise the integrity of IP networks.[22]

Similarly there are now commercial services that provide real-time monitoring or "strategic intelligence" on social media. Three such services are Visible, Gnip, and DataSift. What they do is similar to what was once done by news-clipping services that helped companies manage their "brands." The following is from Twitter's description of its partner firm DataSift:

DataSift is a real-time media curation platform, allowing you to mine the Twitter Firehose for tweets matching the specific criteria of your choice. DataSift's custom Curation Stream Definition Language allows you to filter based on any meta data within a tweet, in addition to a number of other data providers from around the Web. (https://dev.twitter.com/docs/twitter-data-providers)

The Internet Archive has a service used by libraries, called Archive-IT, that can scrape websites regularly for archival purposes. It is also worth noting that such traffic-analysis tools can be used for defensive or protective purposes, such as monitoring outgoing Internet traffic from your machine and alerting you when unauthorized communication (for instance, a rogue application sending your contact list to a server) is occurring.

Because of the amount of information in motion, filters are becoming more and more important. The algorithms that filter Google News and Facebook's newsfeed determine in large part what is newsworthy, what is seen, and what is known by users of Google News and Facebook. There is even evidence that these filters can manipulate users, as Facebook researchers discovered when they experimented with the sentiments of posts in the feeds of selected users. In the infamous "emotional contagion" experiment, they found that by manipulating the number of positive or negative stories readers saw they could manipulate how the readers felt (or, at least, how positive their posts were).[23] As the sociologist Zeynep Tufekci (2014) has pointed out, "these proprietary algorithms determine the visibility of content and can be changed at will, with enormous consequences for political speech."

Big data has been trumpeted as the next big thing made possible by the explosion of digital data about us, to which we contribute. As we write this, big data probably is at the high point in the Hype Cycle[24]—there are inflated expectations as to how it will transform business, government, and the academy, and soon there are likely to be more sober evaluations. Drawing on Mayer-Schönberger and Cukier's book *Big Data*, we characterize how big data changes methods as follows:

**From samples to comprehensive data** In the humanities and the social sciences, traditionally we either focus on works that are considered distinctive or sample larger phenomena. Big data draws on relatively complete datasets, often created for entirely different purposes, and uses them for new research. These datasets are comprehensive in a way that sampling isn't, and in a way that can change the types of questions asked and can change people's confidence in the answers.

**From purposeful data to messy and repurposed data** Traditionally, the data used in research are carefully gathered and curated for that research. With big data, we repurpose messy data from other sources that weren't structured for research and we aggregate data from multiple sources. This raises ethical issues when people's information is used for purposes they didn't consent to; it also means that we can extract novel insights from old data, which is why preservation of data is important.

**From causation to correlation** Big data typically can't be used to prove causal links between phenomena the way experiments can. Instead, big data is used to show correlations. Big data is being pitched as an opportunity to reuse data to explore for new insights in the form of correlations. There are even techniques that can comb through datasets to identify all the statistically significant correlations that could be of research interest.

## What Does This Mean for the Humanities?

What does this mean for the humanities in general and the Digital Humanities in particular? Well, there are obvious privacy issues, ethical issues, and governance issues that call for attention. The liberal arts are supposed to prepare free (*liber*) people for citizenship and participation in democratic community. Big data is now one way citizens are managed both by industry and by government.[25] The liberal arts clearly have a role in studying and teaching about big data. To get a conversation of the phenomenon going, the researchers danah boyd and Kate Crawford (2012) offer six provocative statements:

1. Big Data changes the definition of knowledge.
2. Claims to objectivity and accuracy are misleading.
3. Bigger data are not always better data.
4. Taken out of context, Big Data loses its meaning.
5. Just because it is accessible does not make it ethical.
6. Limited access to Big Data creates new digital divides.

Following boyd and Crawford, one way to think about big data is to look at the pragmatics of how working with large amounts of data changes knowledge. A particularly Digital Humanities way to do this is to think through experimenting with the analysis of large datasets. "Thinking through" is an approach of understanding a phenomenon (thinking about it) through the practices of making, experimenting, and fiddling. It is one of the ways the Digital Humanities can contribute to the larger dialogue of the humanities through what Tito Orlandi and later Willard McCarty called

modeling.[26] We create models of systems and learn through the iterative making and reflecting on the models—especially when they fail, as they usually do. The false positives that our hermeneutica throw up tell us about the limitations of our models and hence about the limitations of the theories of knowledge on which they are based.

## Haven't We Always Been Anxious about Big Data?

There is also a history of thinking through big data, depending on how we define big data. For example, the perception of too much information (for which we need some new technology) goes back to the story of the invention of writing in *Phaedrus* (Plato 1925). One of the interesting aspects of that story is the division of labor imposed by King Thamus on the situation that distinguishes between those who invent and those judge inventions. Socrates proposes that the inventors of a technology are not necessarily the best at judging its utility or potential. The same holds true today: the computer scientists and engineers (not to mention members of the intelligence community) who are building big-data infrastructure systems may not be the best judges of the usefulness or ethical implications of such systems. One gets the feeling that the NSA built the massive data gathering and analyzing systems now being leaked because it could, because there was money, because there was the political will after 9/11, and because they didn't want to be accused of not having tried the next time an attack happens. Now the time has come to learn from the history of attempts to manage excess information documented by Ann Blair (2010) and other scholars. Perhaps there is a role for digital humanists as critics of big data.

The problem of too much data reappeared in 1945 in Vannevar Bush's famous article "As We May Think":

> There is a growing mountain of research. But there is increased evidence that we are being bogged down today as specialization extends. The investigator is staggered by the findings and conclusions of thousands of other workers—conclusions which he cannot find time to grasp, much less to remember, as they appear.

More recently, Clay Shirky argued that the problem is not so much information overload as it is "filter failure."[27] In some cases technologies and tools do exist to filter information, but seldom are they used effectively, as many of us realize when observing the scattering of our browser bookmarks, research notes, bibliographic entries, and other fragments of knowledge. Filtering and management aren't solely technical challenges; they are also

matters of social practices and personal habits. It is worth paying especially close attention to who creates filters, how they work, and how they fail us.

We have seen humanities projects dealing with big data for some time. James Murray and other editors of the *Oxford English Dictionary*, for example, had to deal with millions of quotation slips without any computing in the late nineteenth century and the early twentieth century. Two important early computing projects that don't get the attention they deserve, perhaps because neither of them gathered English texts, are FRANTEXT and Thesaurus Linguae Greacae (TLG).

In the 1970s. FRANTEXT assembled what was then considered an enormous text database of important French texts to provide examples for an extensive linguistic study of the French language.[28] In an 1989 article titled "L'exploitation des grands corpus: Le bestiaire de la littérature française," Etienne Brunet discussed what could be done with a terminal connection to FRANTEXT using the Stella system (mentioned in chapter 5 above). That may not seem like big data now, but in the 1980s, when most of us were lucky to have the full text of a single work or a small collection by a single author, FRANTEXT's 2,500-text corpus was really big, and it provoked Brunet to speculate on and try new methods suited to "grand" collections. Foreseeing the importance of shifting methods, Brunet wrote about a disequilibrium between the breadth of our ambitions and the smallness of the data we typically use, essentially making the case for much larger collections and new statistical and visualization methods. For Brunet, *grand* data allows analysis by comparison; it has built in control samples.

In 1972 the TLG project began gathering all Greek literature between Homer (eighth century BCE) and CE 600, which was then published on CD-ROM. David W. Packard and William Johnson even developed a custom workstation, the Ibycus Scholarly Personal Computer (PSC), in 1985. The PSC could efficiently search the entire TLG and properly display the Greek. In effect, classicists by 1985 had a fairly complete, and constantly improving, collection of everything important written in Greek—a comprehensive dataset to study, rather than samples. This may not have been a big collection by today's petabyte standards, but it was one of the first instances of scholars' being able to ask questions about everything in a domain. The scale of big data is in the eye of the beholder (and the available technologies). The TLG allowed graduate students in Classics to survey every use of a word, something that even today can't be done in most disciplines. Colleagues complained, as did Thamus in Plato's *Phaedrus*, about the effects of this technology on scholarly memory. Who would value the panoptic memory of the experienced scholar when one could search everything on

a workstation? Like today's big data, the size of the collection strained the capabilities of the computing equipment that was available at the time, prompting the development of specialized technology.[29]

### What Can We Do with Big Data Analytics?

At this point, let us take a quick tour through some of the opportunities and risks for Big Data in the Digital Humanities.[30]

**Filtering and subsetting** We begin with a basic operation comparable to what intelligence services do with Information in Motion: filtering big data to get useful subsets that can then be studied by other, possibly traditional techniques (e.g., reading). The Cornell Web Lab, which alas was closed in 2011, was working with the Internet Archive to build a system that would enable social scientists (and humanists) to extract subsets from the archive. The dearth of resources that allow for navigation between large databases of content (among them the Gutenberg Project of digital texts) to specialized environments for reading and analysis should be noted here.

**Enrichment** "Enrichment" is a general term for adding value. In the context of big data, researchers are developing ways to enrich large corpora automatically using the knowledge in big data. In "What Do You Do with a Million Books?" (the 2006 article that first prompted many of us to begin thinking about what big data meant in the humanities), Greg Crane wrote eloquently about what we now can do to enrich big data to make it more useful. Crane has made the further point that simply providing translation enrichment could provide us with a platform for a dialogue among civilizations.

**Sequence alignment** Sequence alignment can be adapted from bioinformatics for the purpose of following passages through time. This is the research equivalent to what plagiarism detectors do with students' papers. They look at sequences of text to see whether similar passages can be found in other texts, contemporary or not. Horton, Olsen, and Roe (2010) described work they had done with the ARTFL textbase to track how passages in an influential twentieth-century history by Paul Hazard had been cribbed from a seventeenth-century text. The point is not that Hazard was a lazy note taker, but that one can follow the expression of ideas across writers when one has them in digital form.

**Diachronic analysis** Diachronic analysis is using data sets to study change over time. Although the use of large diachronic databases dates back to

the Trésor de la langue française (TLF) and the Thesaurus Linguae Grecae (TLG) projects, diachronic analysis got a lot of attention after Google made its Ngram Viewer available and after Jean-Baptiste Michel and colleagues published their paper "On Quantitative Analysis of Culture Using Millions of Digitized Books" (2010). There are dangers to using Ngrams (or phrases of words) to follow ideas when the very words have changed in meaning, orthography, and use over time, but the Ngram Viewer is nonetheless a powerful tool for testing change in language over time (and its search-and-graph approach makes it accessible to a broad public).

**Classification and clustering** Classification and clustering comprise a large family of techniques that go under various names and can be used to explore data. Classification and clustering techniques typically work on large collections of documents and allow us to automatically classify the genre of a work or to ask the computer to generate clusters according to specific features and see if they correspond to existing classifications. A related technique, Topic Modeling, identifies clusters of words that could be the major "topics" (distinctive terms that co-occur) of a large collection. An example was given in the previous chapter, where we used correspondence analysis to identify clusters of key words and years in the Humanist archives.

**Social network analysis** Social network analysis (SNA) involves identifying people and other entities and then analyzing how they are linked in the data. It is popular both in the intelligence community and in the social sciences. SNA techniques can graph a network of people to show how they are connected and to what degree they are connected. In the social sciences, the data underlying a network typically are gathered manually. One might for example interview members of a community in order to understand the network of relationships in that community. The resulting data about the links between people can be visualized or queried by computer. These techniques can be applied in the humanities when one wants to track the connections between characters in a work (Moretti 2013), or the connections between correspondents in a collection of letters or places mentioned in a play. Just as an ethnographer might formally document relationships between people, a historian might be interested in determining who lived in Athens when Socrates was martyred and how those people were connected. Large collections of church records, letters, and other documents now allow us to study social networks of the past. Thanks to a new family of techniques that can recognize the names of people, organizations, or places in a text, it is

now possible to extract named entity data automatically from large text collections, as the Voyant tool called RezoViz does.

**Self-tracking** Self-tracking is the application of big data to your life. You can gather big data about yourself. Personal surveillance technologies allow you to gather lots of data about where you go and who you correspond with, and to then analyze it. The idea is to "know thyself" better and in more detail (or perhaps just to remind you to exercise more). Though results of self-tracking may not seem to be big data relative to the datasets gathered by the NSA, they can accumulate to the point where one must use mining tools to make sense of it. A fascinating early project in life tracking was Lifestream, started in the 1990s as a project at Yale by Eric Freeman and David Gelernter, which introduced interesting interface ideas about how information could be organized according to the chronological stream of life.[31] Gordon Bell has a project at Microsoft called MyLifeBits that has led to a commercially available camera, the Vicon Revue; you hang it around your neck, and it shoots pictures all day to frame bits of your life.[32] Recently Stephen Wolfram, of *Mathematica* fame, has posted an interesting blog essay on what he has learned by analyzing data he gathered about his activities—an ongoing process that, he argues, contributes to his self-awareness.[33] Many different tools are now available to enable the rest of us to gather, analyze, and share data about ourselves—especially data on our physical fitness data. (Among those tools are the fitbit, the Nike+ line,[34] and now the Apple Watch.)

What does all this have to do with academic humanities? Ian Lancashire, a pioneering computing humanist, writes in *Forgetful Muses* about "devising diagnostic software for text analysis in longitudinal studies, not only of creative writing, but of e-mail, diaries, blogs, and even conversation" (2010, p. 218). Lancashire and Graeme Hirst presented research showing that the vocabularies of Agatha Christie and Iris Murdoch shrank dramatically late in their lives. They believe that they can distinguish between healthy aging and aging that shows signs of early Alzheimer's Disease in diachronic collections of texts, and that they have documented an increase in editorial intervention to compensate for Christie's memory loss. In *Forgetful Muses* Lancashire goes further to propose a fascinating humanistic theory of the writing mind that marries cognitive studies and literary studies and that draws on cybernetics. While others might track their physical activity, digital humanists track their cognitive activity. As Mayer-Schönberger and Cukier (2013, 93) put it, "datafication is not just about rendering attitudes

and sentiments into an analyzable form, but human behavior as well." This is a very different form of studying the human, but it is of interest to the humanities nonetheless.

## Problems with Information Overload and Big Data

We close this chapter by looking at the limitations of big data and of its analysis. The discussion of the recent Snowden revelations at the beginning of this chapter highlighted the privacy issues raised by big data and data mining. Two things that often get neglected are whether these techniques work and under what conditions they work well enough to be employed regularly. There is an argument that the mining of surveillance data may not actually work very well. Jeff Jonas and Jim Harper state the problem nicely in a 2006 report titled Effective Counterterrorism and the Limited Role of Predictive Data Mining:

> It would be unfortunate if data mining for terrorism discovery had currency within national security, law enforcement, and technology circles because pursuing this use of data mining would waste taxpayer dollars, needlessly infringe on privacy and civil liberties, and misdirect the valuable time and energy of the men and women in the national security community. (Jonas and Harper 2006, 1)

The problem is that when you analyze a lot of data you get a lot of false hits or false positives. The messier the data, the more false hits you get. The subtler the questions asked of the data, the more nuanced and even misleading the answers are likely to be. When you use Google to search for someone with a common name, such as John Smith, with big data you can get overwhelmingly big and disappointingly irrelevant results—result sets that are too big for you to go through to find the answer, and therefore not useful. A cover story in *Time* talks about the "Pizza Hut" problem: The FBI found itself using valuable human resources following "vague and voluminous" leads from the Stellar Wind system that kept pointing out suspicious phone calls that often turned out to be pizza orders (Gellman 2011).

False positives are particularly problematic in predictive data mining of the sort being done by the NSA, which is using it to try to *predict* who might commit a terrorist attack. As Jonas and Harper point out, when businesses use predictive data mining to identify targets (e.g., pregnant women) for targeted marketing the worst that can happen is that they will send inappropriate junk mail to uninterested people or that they will spook people when their predictions are right. But when *governments* indulge in predictive data mining and generate a high percentage of false positives, innocent

people will be targeted and intelligence resources will be wasted. In order for predictive data mining to prevent terrorism, governments will have to follow up on the predictions and spend valuable investigative human resources checking the leads. Jonas and Harper question whether the human intelligence agents and other resources that check out all the predictions might be better spent on traditional human intelligence activities. They ask whether predictive data mining and the follow-up investigations triggered actually generate better results than traditional counterintelligence. Could the Digital Humanities test the usefulness of predictive data mining on historical datasets in order to improve citizens' understandings of the techniques and their value?

A second problem with analysis of big data is the dependence on correlation. It doesn't tell you about what causes what; it tells you only what correlates with what (Mayer-Schönberger and Cukier 2013, chapter 4). With enough data one can get spurious correlations, as there is always something that has the same statistical profile as the phenomenon you are studying. This is the machine equivalent to apophenia, the human tendency to see patterns everywhere, which is akin to what Umberto Eco explores in *Interpretation and Overinterpretation* (1992).

The dependence on correlations has interesting implications. In 2008, Chris Anderson, the editor in chief of *Wired*, announced "the end of theory," by which he meant the end of scientific theory. He argued that with lots of data we don't have to start with a theory and then gather data to test it. We can now use statistical techniques to explore data for correlations with which to explain the world without theory. Proving causation through the scientific method is now being surpassed by exploratory practices that make big science seem a lot more like the humanities. It remains to be seen how big data might change theorizing in the humanities. Could lots of exploratory results overwhelm or trivialize theory? What new theories are needed for working with lots of data?

A related problem is "model drift" or "concept drift," which is what happens when the data change but the analytical model doesn't. Model drift is one explanation for why the Web service Google Flu Trends fails periodically. Google Flu Trends is based on a model of what people search for when they have the flu, and it doesn't always keep up with how language use changes, especially when there are short-term spikes of interest in flu-related terms because of media attention given to the flu. The fact that people's use of language changes over time is a major problem for humanities projects that use historical data. People don't use certain words today the same way they used those words in the past, and that's a problem for text

mining that depends on words. If you are modeling concepts over time, the change in language has to be accounted for in the model—an accounting that raises a variety of interpretive questions. Does a phrase that you are tracking have the same meaning it had in the past, or has its meaning evolved?

In their 2011 book *Top Secret America*, Dana Priest and William Arkin raise cost-effectiveness questions about the dramatic growth of the intelligence sector in the United States—250 percent between 2001 and 2010 (to the extent that we actually know how much funding goes to intelligence). "In Top Secret America," they write on page 103, "more is often the solution." More recently, in response to claims that NSA surveillance has thwarted as many as 54 threats of terrorism, Justin Elliot and Theodoric Meyer (2013) quote Senator Patrick Leahy from a hearing of the Judiciary Committee: "That's plainly wrong, but we still get it in letters to members of Congress, we get it in statements. These weren't all plots and they weren't all thwarted. The American people are getting left with the inaccurate impression of the effectiveness of NSA programs." As Elliot and Meyer point out, there is little evidence to support the effectiveness of the surveillance, partly because of the secrecy surrounding the NSA and partly because the few cases we can check turn out to not be quite as clear-cut (as they are reported first by the agencies and then by the media). The NSA and its supporters seem to be overstating the value of big data. Could the billions be better spent elsewhere? How can the Digital Humanities team up with data scientists to test claims about data mining, and what ethical issues would arise?

Similar questions about cost effectiveness should be asked, and are being asked, about text analysis and mining in the humanities. Does text mining of large corpora provide real insight? There are two types of questions that we need to keep always in mind in analysis, especially in regard to large-scale analysis:

- Are the methods and their application sound, and can they be tested by alternative methods (or at the very least, are we sufficiently circumspect about the data and methodologies)? In large-scale text analysis or "distant reading," it is easy to misapply an algorithm and generate a false result. It is also easy to overinterpret results. Since at the scale of big data one can't confirm a result by re-reading the sources, it is important to confirm results in various ways. There is a temptation to use the size of the data to legitimize not testing results by reading, but there are always way to test results. Jockers (2013) shows us how one should always be

sceptical of results and develop ways of testing results even if one can't test them by reading.

- Are the results worth the effort? Obviously, people who engage in text analysis and in data mining believe that the results are interesting enough to justify continuing, but at an individual level and at a disciplinary level we need to ask whether these methods are worth the resources. Computational analysis is expensive not so much in terms of computing (of which there is an ample supply thanks to high-performance computing initiatives) as in terms of the human effort required to do it well. Is computational analysis worth the training, the negotiating of resources, and the programming it requires? Individuals at Voyant workshops ask this question in various ways. Also important are the disciplinary discussions taking place in departments and faculties that are deciding whether to hire in the Digital Humanities and whether to run courses in analytics. Hermeneutica can help those departments and faculties to "kick the tires" of text analysis as a pragmatic way of testing analytical methods in general.

We have our own story about too many results—a story that points out another form of false positive in which the falsity is more that of a false friend. In May of 2010 we ran a workshop on High-Performance Computing (HPC) and the Digital Humanities at the University of Alberta.[35] We brought together a number of teams with HPC support so they could prototype HPC applications in the humanities. One team adapted Patrick Juola's idea for a Conjecturator (Juola and Bernola 2009) so that it could run on one of the University of Alberta's HPC machines. The Conjecturator is a data-mining tool that can generate a nearly infinite number of conjectures about a big dataset and then test them to see if they are statistically interesting. The opposite of a "finding" tool that shows only what you ask for, it generates statistically tested assertions about your dataset that could be studied more closely. The assertions take this form:

*Feature **A** appears more/less often*
*in the group of texts **B** than in group **C***
*that are distinguished by structural feature **D***

In the experiment we ran in 2010, we used a corpus of nineteenth-century novels grouped according to the decades in which they were published (e.g., the 1850s). The "features" came from a reverse-engineered English thesaurus. Each "feature" was a set of words around a key word, such as "ground" or "instant." The Conjecturator generated hypotheses such as

"shirt-words appear more often in the novels of the 1850s than in the novels of the 1860s." Feature A (shirt-words) might be all the words from the thesaurus that are related to shirts, group B might be the novels of the 1850s, group C might be the novels of the 1860s, and the distinguishing structural feature D might be decade. The process, once we had adapted it for the WestGrid HPC cluster, ran for about 100 compute-days and generated 87,000 statistically valid assertions! One shudders to think of how many conjectures we would have generated if we had had access to all 1,000 cores of that WestGrid HPC cluster or if we had run the process on a larger corpus.[36]

That experiment raised the new question "What do you do with 87,000 positives?" Well, when you look at the results you find a number of things:

- Many of the results are trivial. They may be statistically valid, but many nonetheless are uninteresting to humans.
- Many of the results are related. You might find that there is a set of related assertions about shirts over many decades.
- The results are hard to interpret in the dialogical sense. You may find a really interesting assertion but have no way to test it, follow it, or unpack it. The Conjecturator is not usefully connected to a full-text-analysis environment in which one could explore an interesting assertion. As always, the trajectory from the very atomistic level (the narrow and specific assertions) to larger, more significant aspects is unmarked and hazardous, though not without potential rewards.

Above all, 87,000 results are too much for anyone to deal with. Such a number is mind-numbing, even demoralizing. It tells us something about the humanity of research that we don't want to be handed so many assertions; we want to find a few all by ourselves. The finding of interesting truths is as important as the truths found. The conjectures above are false positives not in the sense of being untrue, but in the sense that they are false to our way of research. In the digital epoch we console ourselves that at least we are asking the questions, but that isn't true. Computers can ask questions too, we just don't want to hear them. They sound false to us.

Socrates would also remind us that wisdom is not about the quantity of information you possess, or the quantity you can process, but about the thinking that led you to assertions. Information isn't knowledge. Knowledge is held by some person who arrived at that knowledge. Just as owning a book on French grammar isn't the same as knowing French, having 87,000 results isn't the same as understanding nineteenth-century

literature. The possession of big data and big results isn't a certificate of knowledge, and in lazy hands it can become a false positive.

We should therefore listen to the spirit of Plato and recognize that some of the techniques discussed above should be judged false leads and that humanists should take responsibility for experimenting with them, testing them, and judging their wisdom. That's what Edward Snowden did. He judged the activities he was privy to, both as a CIA employee and as a contractor for the NSA. He chose to gather the documents he could gather and to share them publicly. Are we humanists and other curators of information willing to step up to this responsibility, or are we happy leaving it to the engineers who invent new technologies?